DDRL: Domain Distribution Reconstruction Learning for Binary Change Detection in Remote Sensing Images

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***Abstract*— Change detection (CD) aims to identify and locate changes in the same observed surface coverage area across bitemporal images. This technique has widespread applications in urban planning, land use, and disaster damage extraction. Deep learning-based CD methods typically use learnable encoders to map bitemporal images to a common domain distribution space, allowing for the discrimination and localization of change and invariant features. However, due to differences in imaging mechanisms, seasons, and shooting angles, a large number of pseudochanges may easily appear, affecting the accurate recognition of the domain distribution space. In addition, binary CD focuses solely on whether scene targets have changed, resulting in change labels that encompass a variety of different objects, thus increasing the significance of intraclass differences. To address the aforementioned issues, we propose a domain distribution reconstruction learning (DDRL) framework for binary CD, which effectively mitigates the problem of pseudochanges by detecting abnormal feature domain distributions. Specifically, DDRL first extracts multiscale features from bitemporal images using a Siamese cross-window self-attention module, achieving feature domain transformation from the original space. Subsequently, it employs a graph attention enhanced (GAE) module to improve the low-level domain distribution, enabling it to focus on change regions. In addition, DDRL utilizes a cross-domain feature contrastive learning (CFCL) module for reconstructive learning of high-level fused features. This process ensures that intraclass features are compact, while interclass features are dispersed within the high-level domain distribution, thereby significantly improving the domain distribution representation to discriminate pseudochanges. Experimental results show that the proposed DDRL performs excellently across multiple public datasets, surpassing mainstream methods and significantly improving CD performance. The source code will be made available at https://github.com/yzygit1230/DDRL.**

***Index Terms*— Change detection (CD), domain distribution, graph attention enhanced (GAE), reconstruction learning.**

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I. Introduction

CHANGE detection (CD) aims to evaluate changes in various objects within the same area by comparing images taken at different time points and identifying the specific locations of these changes [1], [2], [3]. CD mainly includes three categories: object-oriented CD, binary CD, and multiclass semantic CD. The practicality and convenience of binary CD make it widely applicable to a variety of real-world scenarios, including urban planning, environmental monitoring, and resource management [4], [5], [6]. With the continuous advancement of remote sensing imaging technology, the resolution and detail of remote sensing images have significantly improved. Traditional binary CD methods, which rely on simple pixel differences or change vectors, struggle to capture complex change patterns and can no longer meet the demands of real-world tasks [7], [8]. Consequently, the urgent need has driven researchers to explore more advanced and effective binary CD methods.

Technological advancements have driven the development of CD methods by introducing deep learning-based methods with remarkable capabilities [9], [10], [11]. There are two types of deep learning CD methods: segmentation-based CD methods and metric-based CD methods. As shown in Fig. 1(a), segmentation-based CD methods initiate the process by utilizing a Siamese encoder to extract features from the input bitemporal images, followed by a decoder that integrates features to produce CD results. By employing a segmentation loss function to compare the predicted CD results with the ground-truth change labels, the model parameters are iteratively optimized, enabling the network to learn more robust change features. In addition, the incorporation of advanced techniques such as attention mechanisms [12],

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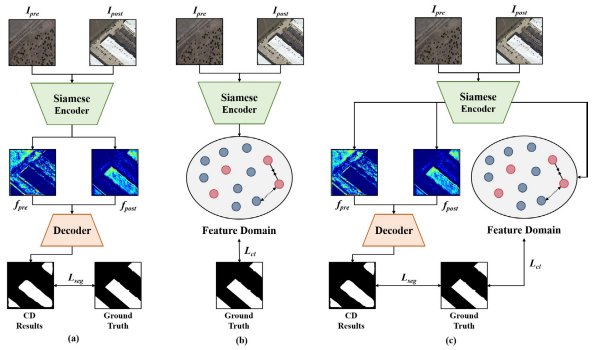


Fig. 1. Architecture of different deep learning CD methods. (a) Segmentation-based CD methods, (b) metric-based CD methods, and (c) domain distribution-based CD framework.

[13], dense feature aggregation [14], [15], and multiscale feature alignment [16], [17] across various network layers significantly enhances the network's capacity to extract change features effectively. It is evident that the performance of segmentation-based CD methods heavily relies on the effectiveness of various modules within the network, with segmentation loss computation occurring only at the deepest layers to optimize parameter learning. In contrast, metric-based CD methods, as illustrated in Fig. 1(b), employ a Siamese encoder to extract different feature vector representations within the original bitemporal image space, encapsulating abstract representations of features at various locations. After comparing features within a feature domain at different time points, distances between various features inside the feature domain are calculated, and pixels with distances exceeding a predefined threshold are identified as change regions. Therefore, the core of metric-based CD methods lies in minimizing the metric distances between time-variant features in invariant regions while maximizing these distances in change regions, ultimately categorizing the domain distribution distance map into change and unchanged categories using distance thresholds. In addition, the integration of generative adversarial networks [18], [19] and optimized contrastive learning losses [20], [21] has been introduced to better learn and differentiate feature distributions across bitemporal images.

Despite the compelling results achieved by the aforementioned two types of deep learning CD methods in various applications, they still exhibit certain limitations. On the one hand, segmentation-based CD methods suffer from insufficient deep feature mining due to the lack of effective multilayer domain feature optimization. Siamese networks employ weight-sharing encoders to extract features from bitemporal images, ensuring that the same objects at different time points are mapped to similar feature space representations. Thus, the same objects at the same locations will have similar feature representations in the feature space. However, most segmentation-based CD methods optimize the model by calculating the loss only from the deepest feature maps generated at the decoder stage, lacking explicit constraints on the learning process of the Siamese network. Consequently, due to the inadequate exploration of semantic relationships across multiple layers, segmentation-based CD methods have limited capacity to learn information with significant intradomain differences. On the other hand, metric-based CD methods transform bitemporal image features into a feature domain using Siamese encoders, employing metric distances to directly optimize the Siamese encoders. By manually setting thresholds, these approaches minimize the metric distances between features in unchanged regions while maximizing those in changed regions, ultimately generating CD results. However, land cover at the same location can differ significantly between bitemporal images, introducing noise and interference into the feature domain, which inevitably leads to false positives and missed detections. Thus, the aforementioned two methods each exhibit distinct strengths and weaknesses. Segmentation-based CD methods excel at directly learning valuable features and change patterns from bitemporal images, thereby reducing the complexity of manual threshold selection. This makes them well-suited to scenarios with well-defined change regions. Conversely, metric-based CD methods delve into the metric relationships between unchanged and changed areas within the feature domain, constraining feature learning during the Siamese encoder stage. This capability is lacking in segmentation-based CD methods, granting metric-based CD methods a notable advantage in handling complex backgrounds and subtle changes. Therefore, it is worthwhile to explore the effective integration of the complementary advantages between the above two CD methods.

It is well known that although homologous remote sensing images are all optical images, differences in imaging mechanisms, seasons, and shooting angles can easily lead to a large amount of pseudochange interference. This interference makes it difficult to discern subtle differences between features within the feature domain. To this end, we propose a domain distribution reconstruction learning (DDRL) framework for CD. As shown in Fig. 1(c), DDRL seamlessly integrates the strengths of both segmentation-based CD methods and metric-based CD methods, forming a domain distribution-based CD framework. After implementing feature domain transformations via the Siamese encoder, it utilizes the principles of metric-based CD methods to perform reconstruction learning on the extracted multiscale features within the feature domain. This constrains the Siamese encoder to effectively reconstruct the abnormal feature domain, guiding the decoder to focus more accurately on change regions. In addition, by comparing the CD results with ground-truth labels, the network directly learns meaningful change patterns from the bitemporal images. This implicit and explicit co-optimization of the network synergistically enhances its ability to perceive change regions while mitigating the interference caused by pseudochanges. The main contributions of this work include the following.

1) We propose a DDRL framework for CD that utilizes reconstruction learning to compact intraclass features and disperse interclass features within the feature domain, enhancing the perception of different category features and accurately identifying change regions in bitemporal images. To the best of our knowledge, DDRL represents an innovative endeavor that combines the strengths of both segmentation-based CD methods and metric-based CD methods.

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2) We design a graph attention enhanced (GAE) module to optimize low-level domain distributions, which iteratively propagates and aggregates feature information across the graph, emphasizing significant changes in object features and reducing the influence of irrelevant noise. Thus, the network is able to perceive changed regions and provides prior information for subsequent deep feature extraction.
3) We construct a cross-domain feature contrastive learning (CFCL) module that employs depth contrastive learning loss to perform reconstruction learning on depth-fused features. This module explores global semantic relationships among various features, reducing the metric distance between unchanged features while increasing the distance between changed features. Besides, we develop a significant feature region repository that stores representative embeddings, enhancing the ability to discriminate pseudochanges.

II. Related Work

A. Segmentation-Based CD

Traditional image segmentation typically uses a single image as input, whereas CD involves bitemporal images of the same location taken at different times. Early segmentation-based CD methods either perform image differencing or concatenation for CD input, which inevitably leads to the loss of some detail information [22]. Subsequently, some studies have used weight-sharing Siamese networks to process bitemporal images, which improves the accuracy of feature comparison, permitting better detection of changing locations and categories. For example, GASNet [23] utilizes a global perception Siamese network to enhance the representation of change features by incorporating co-occurrence relationships between background and change regions, thus generating effective global CD results. CARGNet [24] leverages the point-based changed regional growth module to transform sparse point labels into comprehensive expanded labels. The base decoder and the extended decoder are supervised by point labels and expanded labels, respectively. Finally, consistency in CD results is achieved by minimizing the difference between the predictions of the two decoders. SAGNet [25] employs a Siamese encoder to extract highly representative features and employs a bilateral feature fusion module to effectively integrate differential and global information, thereby improving the localization of change details. SMDNet [26] utilizes a Siamese feature difference encoder to capture differences between images while perceiving the edge features of objects. Then, it integrates key features using progressive sampling to enhance adaptability to different environments. Although these efforts have improved the accuracy of CD to some extent, segmentation-based CD methods still fall short of fully learning change information. Since they only output CD results at the deepest layer of the network and compare them with change labels to learn change patterns, this implicit learning approach fails to fully exploit the potential of shallow features. Besides, the lack of explicit constraints during feature extraction results in deep features containing a significant amount of redundant information and noise.

B. Metric-Based CD

Different from segmentation-based CD methods, metric-based CD methods adopt an explicit approach to optimize feature similarity between bitemporal images in the feature domain. The motivation for applying the idea of domain distribution contrastive learning to CD stems from the observation that when features extracted by the Siamese encoder are mapped into the feature domain, the feature vectors of changed regions exhibit significant dissimilarity, while those of unchanged regions show similarity to each other [27]. Therefore, by explicitly measuring and optimizing the similarity between features of unchanged regions and the dissimilarity between features of changed regions through contrastive learning loss, metric-based methods can achieve more precise CD. DASNet [28] enhances feature discriminability within the feature domain by employing a dual attention module. Furthermore, it utilizes a weighted double-margin contrastive loss to optimize the distance between features of different classes, effectively addressing class imbalance issues. DSAM-Net [29] employs a deep metric learning approach, using a metric module to learn the metric distance of intermediate layer embedded feature vectors from the feature domain. This provides more discriminative features, thereby facilitating the learning of change mappings. Xu et al. [30] utilize capsule networks to extract vector representations from bitemporal images. By leveraging vector cosine similarity and vector difference, they conducted feature comparisons within the capsule network, effectively increasing the discriminability between changed and unchanged pixels. CDSCL [31] builds upon the Siamese contrastive learning structure by employing a negative cosine similarity contrastive loss function. This approach ensures a tighter interaction between the features extracted by the Siamese encoders, resulting in more consistent CD results. Although metric-based CD methods offer stronger interpretability in feature representation, the presence of numerous pseudochanges in temporal images inevitably impacts accurate identification in the domain distribution space. Therefore, a potential direction is to combine the complementary advantages of segmentation-based CD methods and metric-based CD methods, which is significant for enhancing the overall performance of CD.

C. Attention Mechanisms for CD

Although Siamese encoders can extract multiscale features rich in semantic information from bitemporal images, they inevitably capture a substantial amount of data from the whole image, which includes some redundant noise information. Inspired by human visual mechanisms, where attention is not uniformly distributed but focused on areas of interest, attention mechanisms have been introduced into CD methods to enable the network to selectively focus on key regional features [32]. For example, PGCFN [33] initially encodes the geospatial position of bitemporal images and then integrates the encoded position with a location-aware graph attention network, resulting in an outstanding performance in urban multiclass CD. CDNeXt [34] introduces the temporal–spatial interaction attention module, which utilizes temporal and spatial attention scores derived from querying the Siamese encoder to

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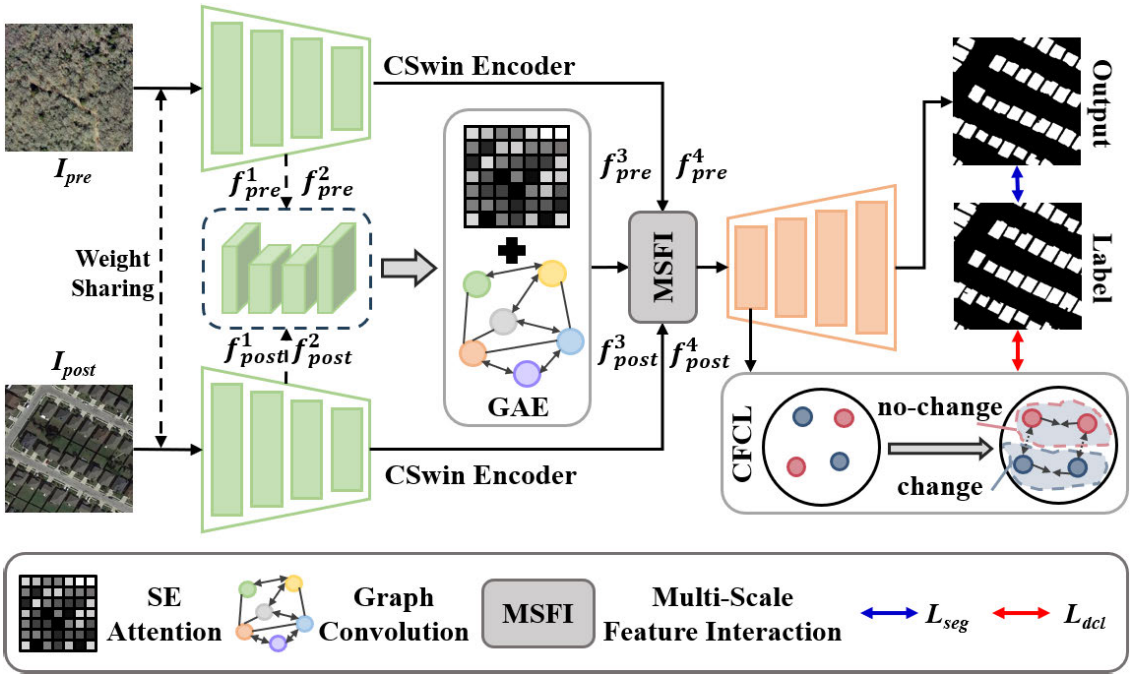


Fig. 2. Overview of the proposed DDRL framework. Overview of the proposed DDRL framework. The bitemporal image is fed to the Siamese encoder to extract multiscale features. The low-level features undergo a GAE module to enhance the feature representation and subsequently perform cross-layer interactions together with the high-level features to generate CD results. Besides, the features obtained from the interaction between high-level features are used to compute the depth contrastive loss to achieve intraclass feature aggregation and interclass feature dispersion, helping the CD results align more closely with the ground-truth labels.

reconstruct spatiotemporal semantic correlations, effectively addressing the impacts of temporal style differences. Although the aforementioned attention mechanisms effectively enhance the focus on local salient information within images, CD tasks involving bitemporal images necessitate a broader and more comprehensive perspective to encapsulate the entirety of changes and their contextual nuances. Simple attention mechanisms may encounter challenges in effectively managing the multiscale features of changes or temporal patterns across diverse time points. Therefore, enhancing the accuracy of CD may require more intricate and adaptive attention mechanisms capable of handling multiscale phenomena, long-term dependencies, and temporal variations in feature representation.

On the basis of the above considerations, significant improvement and optimization can be achieved by combining the advantages of segmentation-based CD methods and metric-based CD methods, explicitly and implicitly optimizing the CD network, and significantly enhancing change features using attention mechanisms.

III. Methodology

A. Overview of DDRL

It is evident that both segmentation- and metric-based CD methods have inherent flaws due to factors such as their detection mechanisms. The former lacks explicit constraints on the loss learning process, while the latter is susceptible to unnecessary interference from metric thresholds. To this end, we propose the DDRL framework based on DDRL, which effectively overcomes the shortcomings of these two methods.

As shown in Fig. 2, our proposed DDRL comprises a Siamese cross-window self-attention encoder, a GAE module for enhancing low-level features, an MSFI module for feature interaction, and a CDCL module based on metric and contrastive learning. Initially, the original bitemporal images I_{pre} and I_{post} are fed into the Siamese cross-window self-attention encoder to extract multiscale features $(I^i_{\text{pre}}, I^i_{\text{post}})$, $i \in [1, 4]$. The low-level features $(I^i_{\text{pre}}, I^i_{\text{post}})$, $i \in [1, 2]$, after being enhanced in the channel dimension by the GAE module, are concatenated with the high-level features $(I^i_{\text{pre}}, I^i_{\text{post}})$, $i \in [3, 4]$, and fed into the multiscale feature interaction (MSFI) module for layer-by-layer interaction, producing CD results through the decoder. In terms of loss constraints, DDRL calculates segmentation loss using CD results and ground-truth labels. Simultaneously, within the CFCL module, high-level fused features and ground-truth labels are used to achieve intraclass feature aggregation and interclass feature dispersion, imposing global constraints on the embedding space based on pixel-to-pixel similarity.

B. Siamese Cross-Window Self-Attention Module

The Siamese encoder-based network, which is the current mainstream CD framework, effectively learns abstract features and representations from bitemporal images, enabling feature transfer across different domains. The purpose of the Siamese encoder is to extract discriminative feature space representations from the abnormal regions in the bitemporal images while maintaining consistent feature representations for the unchanged regions. Therefore, the feature extraction capability of the Siamese encoder largely determines the overall CD performance of the model. The state-of-the-art backbone, CSwin [35], has shown excellent performance in the field of computer vision. To fully integrate the Siamese encoder with subsequent domain distribution processing modules, we employ CSwin as the Siamese cross-window self-attention

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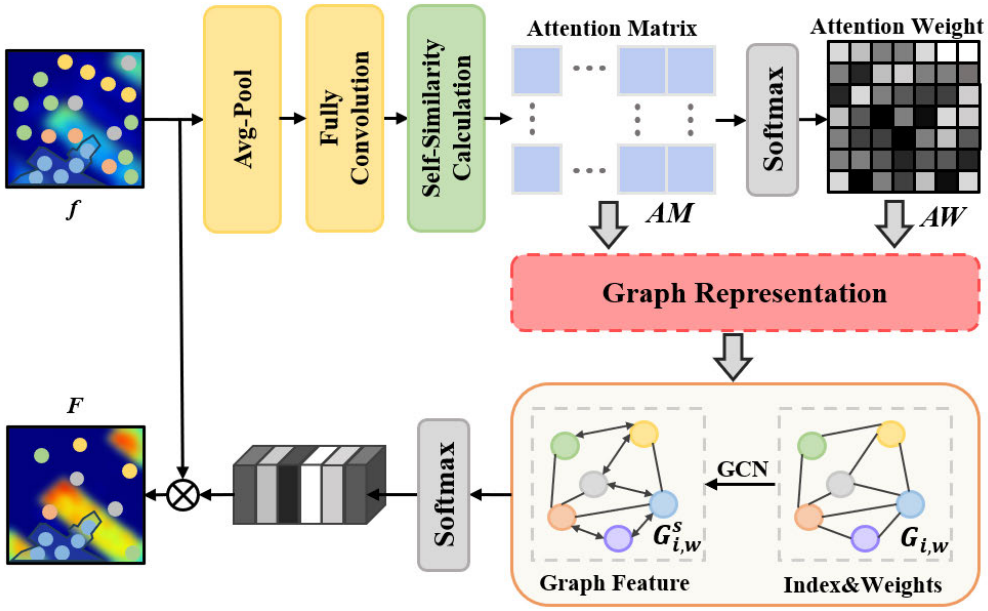


Fig. 3. Overview of the proposed graph attention enhanced module.

module for DDRL. Benefiting from window attention and locally enhanced positional encoding structures, the Siamese cross-window self-attention module adapts to input features of different sizes and is highly suitable for downstream CD tasks. We use the bitemporal multiscale features processed through its four attention stages as the output of the Siamese encoder, which is then applied to the subsequent GAE and CFCL modules.

C. GAE Module

In bitemporal images, the severe class imbalance leads to a significant gradient vanishing problem since the channel attention mechanism weighs multiscale features to focus on key channel features, and graph convolution captures structural dependencies between key feature nodes through domain aggregation. Therefore, as shown in Fig. 3, we employ the GAE module to enhance the low-level domain distribution in the channel dimension, allowing it to focus on change areas and reduce the impact of irrelevant noise, thereby effectively representing the features of the change regions.

Specifically, we execute graph-based relationship-aware operations in the channel dimension to construct more comprehensive context dependencies. The original bitemporal images are processed through the Siamese cross-window self-attention module to obtain multiscale feature maps. Since low-level features contain more spatial information and the computation of high-dimensional feature channels consumes more resources, we only apply graph attention enhancement to the bitemporal features in the first two stages. First, the feature map is fed into a channel processing function $\sigma()$ composed of average pooling and a fully connected layer, which compresses and maps them into feature representation vectors. Subsequently, a self-similarity calculation function $\alpha()$ transforms these feature representation vectors into a channel attention matrix AM, followed by a softmax operation to obtain channel attention weights AW. Therefore, the implementation process of channel attention for the input features can be described as follows:

$$\mathrm{AM} = \sigma(\alpha(f)) \tag{1}$$

$$\mathrm{AW} = \mathrm{softmax}(\mathrm{AM}). \tag{2}$$

Next, we learn the graph relationship between the feature domain distributions of AM and AW in each channel, converting the feature vectors into graph result data and reconstructing the dependencies among graph nodes, edge indices, and edge weights. Specifically, the node features represented by each channel are obtained by a global pooling operation combined with the channel attention weights, and the edges between the nodes are used as adjacency matrices using the channel attention matrices as adjacency matrices for the weight relationships of the edges scheduled between the nodes

$$G_{i,w} = \{G_i, G_w\} = \begin{cases} G_i = \mathrm{AW} \times \mathrm{GAP}(f_{\mathrm{post}}) \\ G_w = \mathrm{AM} \end{cases} \tag{3}$$

where GAP() denotes the global average pooling operation and $G_{i,w}$ denotes the set of edge indices and edge weights of the input features in the upper graph of the corresponding channel. Subsequently, a graph convolution operation module composed of ReLU activation functions is used to perceive the node relationships in $G_{i,w}$, focusing on the change regions and enhancing the weight proportions between the graph nodes, thereby strengthening the representation capability of the change region nodes. The above processes can be calculated as follows:

$$G^s_{i,w} = \mathrm{GCN}(G_{i,w}) \tag{4}$$

where $G^s_{i,w}$ represents the enhanced graph features. Finally, we map $G^s_{i,w}$ into feature vectors using the softmax activation function and multiply them by the input feature f to obtain the output data F of the GAE module

$$F = \mathrm{softmax}\left(G^s_{i,w}\right) \times f. \tag{5}$$

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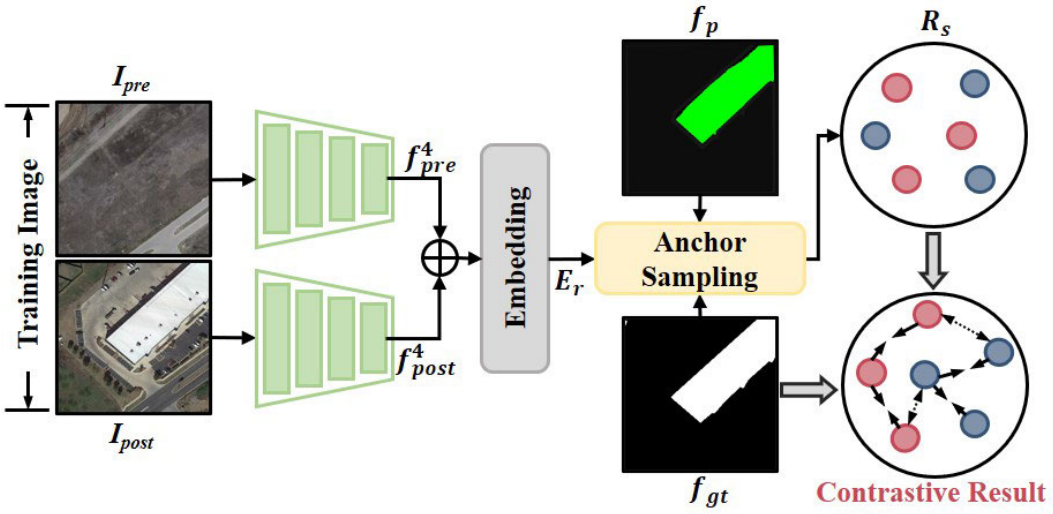


Fig. 4. Overview of the proposed CFCL module.

Consequently, the GAE Module leads to more accurate and robust feature representations, significantly improving the ability to recognize and differentiate between significant features by leveraging graph-based attention mechanisms.

D. CFCL Module

Due to the ability of contrastive learning to enhance spatial consistency among similar regions by controlling the similarity between positive and negative sample pairs, it effectively brings together the feature nodes of real change areas. As shown in Fig. 4, our proposed CFCL module primarily involves constructing a significant feature region repository using anchor sampling methods to achieve global semantic awareness of domain distribution. In addition, it utilizes depth contrastive loss to push apart the representation vectors of dissimilar feature regions while pulling closer the representation vectors of similar feature regions.

1) Significant Feature Region Repository: The core idea of the domain distribution contrastive learning proposed in this article is to distinguish between similar and dissimilar features in the projected embedding space, given an anchor point. Considering that deep features contain rich semantic information capable of capturing complex patterns and high-level concepts in the data, the CFCL module concatenates high-level features f^4_{pre} and f^4_{post} extracted by the Siamese encoder to construct the repository. These concatenated features are then mapped into a lower dimensional vector space through a patch embedding operation to generate the embedding E_r. The above processes can be calculated as follows:

$$E_r = E\left(\text{concat}\left(f^4_{\text{pre}}, f^4_{\text{post}}\right)\right) \tag{6}$$

where concat() represents the concatenation operation and $E()$ denotes the patch embedding operation. The significant feature region repository aids the depth contrastive loss by focusing on learning features of the change regions, thereby enhancing their discriminative ability in the reconstruction learning process. We input the embedding E_r, label map f_{gt}, and prediction result f_p into the anchor sampling function $\delta()$ to establish the repository R_s

$$R_s = \delta(E_r, f_{gt}, f_p) \tag{7}$$

where E_r is used to calculate feature differences in the domain distribution, f_{gt} serves as the supervision signal during the training process, and f_p provides initial recognition information of the input images, guiding the selection of significant feature regions. Specifically, the anchor sampling function consists of two main steps: determining significant feature regions and randomly permuting anchor points for sampling. First, we filter the domain features of E_r based on f_p, identifying and counting feature distribution points that exceed the threshold frequency. These points are compared with f_{gt} to determine the indices of significant and nonsignificant regions as anchor points. Next, the sampling regions are dynamically adjusted according to the maximum storage limit and the threshold for significant region storage. Finally, a random permutation is used to establish the final significant feature region repository R_s.

2) Depth Contrastive Loss: The establishment of the repository enables the differentiation of global semantic information, setting preconditions for cross-domain contrastive learning of deep features. This process prompts the CFCL module to focus on significant change regions. We begin by calculating the similarity matrix based on the dot product between the input feature vectors. To avoid numerical overflow, we then apply numerical stabilization to the resulting matrix $\text{sim}(f_i, f_j)$. The above processes can be calculated as follows:

$$\text{sim}(f_i, f_j) = \frac{f_i \times f_j}{\tau} - \max_k \left(\frac{f_i \times f_k}{\tau}\right) \tag{8}$$

where f_i denotes the ith anchor feature from R_s; f_j represents the j-th feature from the labels; τ is the temperature parameter controlling the similarity, which is set to 0.1 in our implementation; and k indicates the column with the highest value in the i-th row, excluding the j-th column. Subsequently, we compute the log probability of the numerically stabilized similarity matrix to measure the similarity between change and

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nonchange features within the domain distribution area

$$\log P(f_i, f_j) = \text{sim}(f_i, f_j) - \log\left(\sum_k \exp(\text{sim}(f_i, f_j))\right) \tag{9}$$

where exp() and log() denote the exponential and logarithmic operations, respectively. Finally, we utilize a label mask to perform a weighted average of the log probabilities for each feature vector, thereby optimizing the model's feature representation. The depth contrastive loss can be formulated as follows:

$$L_{\text{dcl}} = -\frac{1}{N}\sum_i \frac{\sum_j M_{ij} \times \log P(f_i, f_j)}{\sum_j M_{ij}} \tag{10}$$

where M_{ij} represents the label mask, indicating whether i and j belong to the same change region, and N denotes the total number of feature indices. Through the comparison of the repository and the label map, the CFCL module reduces the metric distance between invariant features while increasing the metric distance between change features.

E. Objective Function

To guide DDRL in end-to-end feature learning, we employ a hybrid loss function L_{hybrid} for fully supervised training of the CD task as follows:

$$L_{\text{hybrid}}(P, G) = L_{\text{seg}}(P, G) + \lambda L_{\text{dcl}}(P, G) \tag{11}$$

where P represents the deepest prediction CD result of DDRL; G denotes the ground truths; $L_{\text{seg}}()$ is the segmentation loss, which consists of binary cross-entropy loss and Dice loss; $L_{\text{dcl}}()$ is the domain distribution loss; and λ is a custom parameter, which is set to 0.2 in our implementation. The binary cross-entropy loss and the Dice loss can be formulated as follows:

$$L_{\text{BCE}}(P, G) = G\log(P) + (1 - G)\log(1 - P) \tag{12}$$

$$L_{\text{Dice}}(P, G) = 1 - \frac{2 * (P \cap G)}{P + G} \tag{13}$$

where $\cap$ denotes the intersection operation. In (11), $L_{\text{seg}}()$ and $L_{\text{dcl}}()$ complement each other. On the one hand, $L_{\text{dcl}}()$ optimizes DDRL in the metric space, enhancing feature representation and aiding the decoder network in accurately identifying change regions. On the other hand, $L_{\text{seg}}()$ automatically aligns with the CD ground truth, thereby mitigating issues related to pseudochanges and manual intervention.

IV. Experiments

A. Datasets

1) LEVIR-CD: The LEVIR-CD [36] is derived from building change data across multiple cities in Texas, USA, spanning from 2002 to 2018. It includes 637 pairs of remote sensing images with a resolution of 1024 × 1024 pixels and a spatial resolution of 0.5 m. LEVIR-CD encompasses a variety of building change types, including villas, apartments, and large warehouses. We have partitioned the original images into segments of 256 × 256 pixels, removed overlapping regions, and split them into training, validation, and test sets, resulting in 3096, 432, and 921 pairs of bitemporal images and their corresponding labels, respectively.

2) WHU-CD: The WHU-CD [37] is released by Wuhan University, which includes land cover change data from the post-earthquake reconstruction in Christchurch, New Zealand, spanning from 2012 to 2016. It contains a pair of remote sensing images with a size of 32 507 × 15 354 pixels and a spatial resolution of 0.75 m. We have segmented the original images into nonoverlapping image patches of 256 × 256 pixels. We further divided the image patches into a training set with 5201 images, a validation set with 744 images, and a test set with 1487 images.

3) SYSU-CD: The SYSU-CD [29] is released by Sun Yat-sen University, which includes land cover change data from Hong Kong spanning from 2007 to 2014. It comprises 20 000 pairs of VHR remote sensing images, each with a size of 256 × 256 pixels and a spatial resolution of 0.5 m. This dataset covers a variety of complex change types such as buildings, urban roads, ports and ships, and mountainous vegetation, with significant interference from pseudochanges. We further divided the image patches into a training set containing 12 000 images, a validation set with 4000 images, and a test set comprising 4000 images.

TABLE I
Quantitative Comparison Results on the LEVIR-CD. Optimal Values Are Shown in **Bold**

Method	Precision(%)	Recall(%)	F1(%)	mIOU(%)	Kappa(%)
FC-EF [38]	79.91	82.84	81.35	81.97	78.99
LUNet [39]	85.69	90.99	88.73	88.44	87.28
STANet [36]	82.73	**93.84**	87.93	87.65	86.33
IFNet [40]	85.37	90.24	87.74	87.53	86.17
BITNet [41]	87.32	91.41	89.32	89.00	87.96
HFANet [42]	83.36	91.47	87.23	87.03	85.56
DPCCNet [43]	87.93	92.64	90.22	89.86	88.97
WNet [44]	89.73	89.81	89.77	89.47	88.50
CSINet [45]	88.61	93.61	91.04	90.06	89.89
DDRL	**93.54**	91.31	**92.41**	**92.02**	**91.48**

B. Experimental Settings

We conduct all experiments using the PyTorch framework on the NVIDIA A100 GPU with memory of 80 GB. We train all models using the same experimental parameters, including a batch size of 32. All methods are performed for 500 epochs of training; the highest weights for F1 indicators in the validation set are used as tests. Besides, we employ AdamW to update the gradient, with an initial learning rate of 0.00035. We use five evaluation metrics to evaluate CD methods, including precision, recall, F1, mIOU, and kappa coefficient [34].

C. Comparative Results

1) Comparison Methods: To perform a holistic assessment of the proposed DDRL, we conduct both quantitative and visual analyses using several advanced CD methods, including FC-EF [38], LUNet [39], STANet [36], IFNet [40], BITNet [41], HFANet [42], DPCCNet [43], WNet [44], and CSINet [45].

2) Quantitative Analysis: Table I presents the quantitative metrics results of DDRL compared with other methods on LEVIR-CD. The results demonstrate that our proposed

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TABLE II
QUANTITATIVE COMPARISON RESULTS ON THE WHU-CD. OPTIMAL VALUES ARE SHOWN IN **BOLD**

Method	Precision(%)	Recall(%)	F1(%)	mIOU(%)	Kappa(%)
FC-EF [38]	70.43	92.31	79.90	82.12	78.77
LUNet [39]	66.32	93.06	77.45	80.26	76.13
STANet [36]	76.82	**95.45**	85.43	86.23	84.31
IFNet [40]	86.51	87.69	87.09	87.93	86.45
BITNet [41]	82.35	92.59	87.17	87.96	86.50
HFANet [42]	74.19	89.56	81.15	83.12	80.13
DPCCNet [43]	81.83	93.71	87.37	88.12	86.70
WNet [44]	71.50	94.98	81.59	83.40	80.55
CSINet [45]	86.28	91.51	88.82	89.38	88.25
DDRL	**92.30**	90.58	**91.43**	**91.70**	**91.02**

TABLE III
QUANTITATIVE COMPARISON RESULTS ON THE SYSU-CD. OPTIMAL VALUES ARE SHOWN IN **BOLD**

Method	Precision(%)	Recall(%)	F1(%)	mIOU(%)	Kappa(%)
FC-EF [38]	64.58	**89.79**	75.13	71.19	65.73
LUNet [39]	76.14	81.74	78.84	76.13	72.01
STANet [36]	78.02	81.22	79.59	76.97	73.12
IFNet [40]	75.29	87.60	80.98	77.91	74.52
BITNet [41]	80.61	79.29	79.95	77.53	73.83
HFANet [42]	73.66	85.00	78.92	75.90	71.80
DPCCNet [43]	85.07	75.91	80.23	78.12	74.58
WNet [44]	80.57	81.73	81.14	78.59	75.27
CSINet [45]	77.89	83.02	80.07	77.65	74.06
DDRL	**87.09**	78.16	**82.38**	**80.19**	**77.32**

domain distribution with intraclass aggregation and interclass dispersion constraint mechanism significantly outperforms current mainstream binary CD methods. Specifically, DDRL shows substantial improvements in precision, F1, mIOU, and kappa metrics, with increases of 3.81%, 1.37%, 1.96%, and 1.59%, respectively. Notably, CSINet and DPCCNet exhibit impressive recall values, reaching 93.61% and 92.64%, respectively. Accordingly, CSINet shows enhanced sensitivity to building change areas through an effective fusion of spatiotemporal–spectral features. However, CSINet and DPCCNet lag behind DDRL in the primary evaluation metrics of F1 and precision, suggesting a higher false detection probability for change regions. Overall, the quantitative metrics on the LEVIR-CD confirm that the proposed feature domain distribution CD strategy significantly improves CD performance.

Table II presents the quantitative metrics results of DDRL compared with other methods on the WHU-CD. Evidently, DDRL exhibits similar outstanding performance as observed on the LEVIR-CD, as the change samples in these datasets possess similar characteristic features. This further validates the superior generalization capabilities of DDRL. DDRL achieves a precision of 92.30%, representing a 5.79% improvement over the second-best method, IFNet. Moreover, DDRL surpasses the current best-performing method, the cascade graph convolution-based CSINet, with improvements of 2.16%, 2.32%, and 2.77% in F1, mIOU, and kappa metrics, respectively. Notably, WNet, benefiting from the progressive integration of multilevel difference feature maps and effectively considering local, global, and cross-scale fusion information, achieves an excellent recall of 94.98%. On the other hand, our F1 score reaches 91.43%, which is significantly higher than the 81.59% of WNet. This indicates that our method achieves a favorable tradeoff between precision and recall metrics, resulting in the most accurate detection of change areas.

Table III presents the quantitative metric results of DDRL compared with other methods on the SYSU-CD. The complex variations in land cover types and imaging differences in the SYSU-CD pose significant challenges, resulting in reduced quantitative metrics for all compared methods. Despite these challenges, DDRL achieves the highest scores in precision, F1, mIOU, and kappa on the SYSU-CD. Compared to the second-best performing method, WNet, DDRL improves F1 and mIOU by 1.24% and 1.6%, respectively. In addition, we observe that FC-EF achieves the best recall but performs the worst in other metrics, indicating that such polarized results fail to demonstrate comprehensive CD performance. Therefore, the results in Table III indicate that the combination of enhancing low-level domain feature distribution and integrating high-level domain contrast features in DDRL effectively strengthens the localization of both the main change regions and their boundaries while also enhancing independent cross-domain feature learning. This approach effectively distinguishes similar cross-domain features, mitigating the challenge of pseudochanges.

3) Visual Analysis: Fig. 5 presents the visual results of DDRL compared with other methods in five typical change scenarios from the LEVIR-CD, where red denotes false positives and green denotes false negatives. For the large building change types shown in Fig. 5(a) and (b), FC-EF, IFNet, and BITNet exhibit significant omissions in the main change regions and more false detections along the edges of the change areas. In contrast, DDRL shows only small areas of red and green, indicating that the Siamese cross-window attention mechanism effectively extracts multiscale features with strong representational capacity. In the sparse and irregular change area detection results shown in Fig. 5(c), all comparison methods fail to effectively distinguish cross-class semantic information, resulting in significant omission areas. DDRL, on the other hand, utilizes the GAE module to aggregate intraclass features in the low-level feature domain, achieving precise perception of low-level discriminative feature contextual information. In addition, in the cases of small and densely packed building changes shown in Fig. 5(d) and (e), DPCCNet, CSINet, and DDRL perform better, with relatively more complete edges in the dense and minute change areas. Overall, the visual comparison results on the LEVIR-CD demonstrate that DDRL can effectively detect various change types with high accuracy.

Fig. 6 presents the visual results of DDRL compared with other methods in five typical change scenarios from the WHU-CD. In the large warehouse change type as shown in Fig. 6(a), LUNet mistakenly identifies white markings in nonchange areas as change regions, and all comparison methods show false detections in the nonchange area at the top left corner. Similar false detections and omissions are evident in Fig. 6(b)–(e). HFANet, IFNet, and BITNet struggle to effectively handle the challenge posed by highly similar surface information present in the change regions, leading to prominent red and green pixel areas in the detection results. Methods such as FC-EF, DPCCNet, WNet, and CSINet also perform poorly under the influence of pseudochanges.

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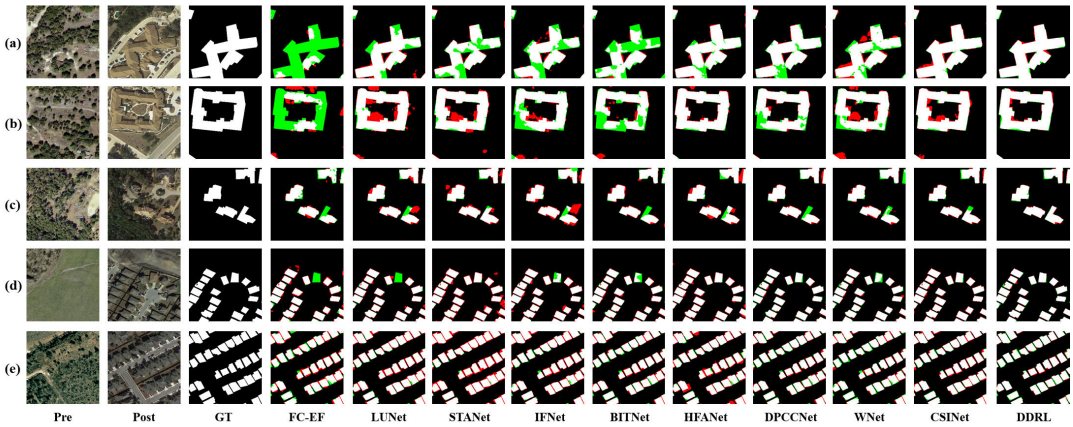


Fig. 5. Visual results of five typical scenes from the LEVIR-CD.

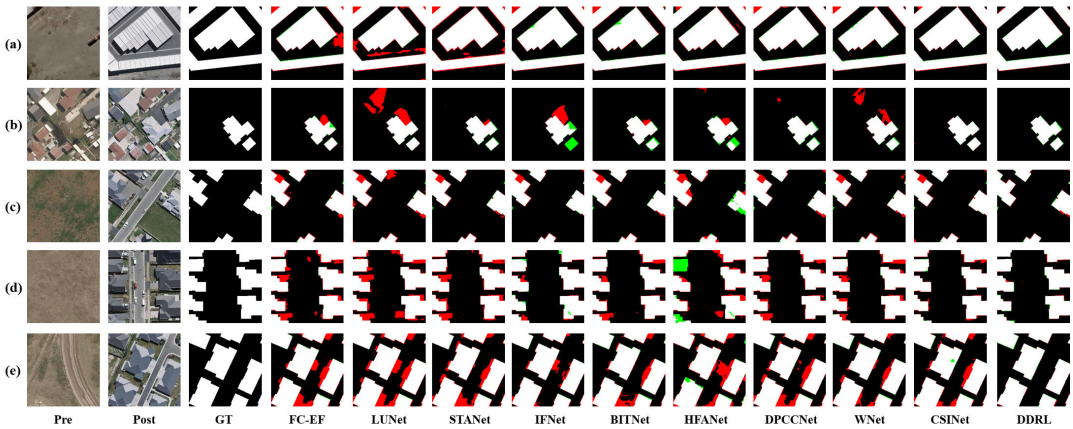


Fig. 6. Visual results of five typical scenes from the WHU-CD.

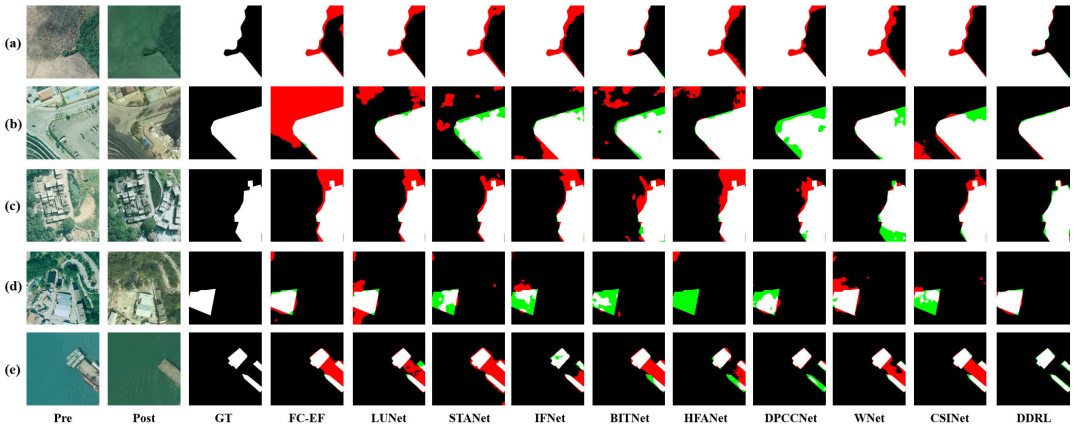


Fig. 7. Visual results of five typical scenes from the SYSU-CD.

In contrast, DDRL demonstrates more accurate details in the change subjects, smoother boundary information, and fewer false detections. As a result, EGCN enhances low-level domain distribution and, when combined with CDCL to reconstruct the consistency of high-level fused features, effectively distinguishes discriminative features within both high- and low-level domains. Consequently, the above modules significantly improve the ability of the DDRL to detect pseudochanging areas.

Fig. 7 illustrates the visual results of DDRL compared with other methods in five typical change scenarios from the SYSU-CD. These scenarios encompass various change types, including vegetation, roads, buildings, and ships, effectively validating the generalization capability of each CD method. As shown in Fig. 7(a), DDRL accurately identifies vegetation change areas with fewer false detections. In Fig. 7(b) and (c), the similar texture between change and nonchange areas introduces uncertainty to CD, and the presence of large shadow areas around the change regions leads to varying degrees of deviations in detection results. DDRL mitigates the impact of irrelevant backgrounds on CD accuracy through a Siamese cross-window attention mechanism for feature domain

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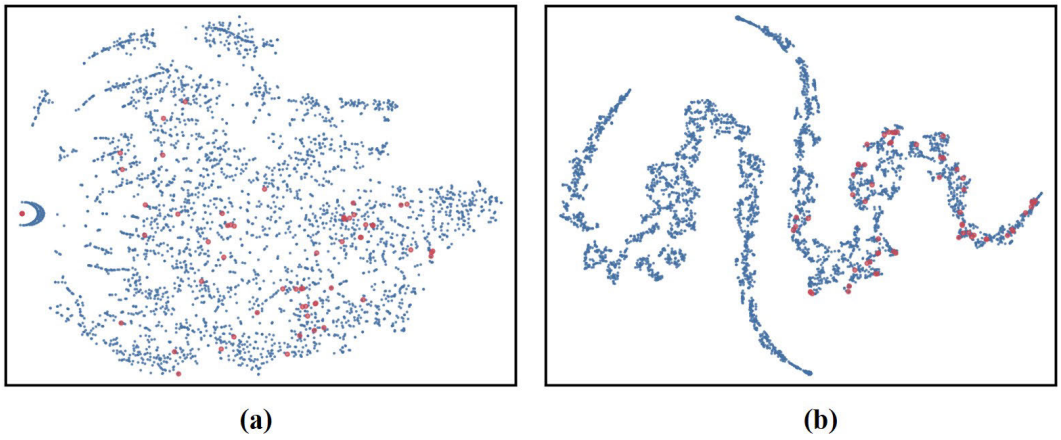


Fig. 8. t-SNE visualization of the DDRL. (a) Without the CFCL module. (b) With the CFCL module.

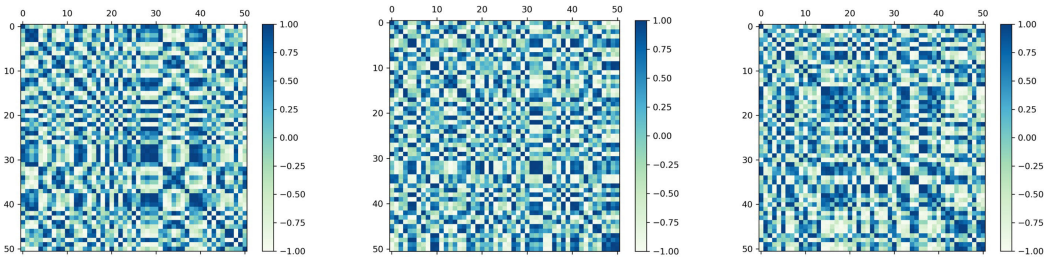


Fig. 9. Correlation matrices of the DDRL from (a) LEVIR-CD, (b) WHU-CD, and (c) SYSU-CD.

transformation. In Fig. 7(d), building demolitions severely disrupt the detection results of most methods, with HFANet showing complete omissions and false detections. In addition, the type of ship change in Fig. 7(e) poses significant challenges to all methods except DPCCNet and DDRL. While DPCCNet effectively identifies unchanged docks, it is less sensitive to changing ship features. DDRL ensures cross-domain feature independence by aggregating intraclass features and dispersing interclass features, thereby adapting to changing scenarios and reducing interference caused by complex environments.

D. Interpretability Analysis

As shown in Fig. 8, we use t-SNE [46] to reduce the dimensionality of the features to two dimensions to assess the discriminative performance of the CFCL module within the domain distributions. Blue represents unchanged feature points, and red represents changed feature points. Due to the significant sample imbalance in CD tasks, where the number of blue points far exceeds that of red points, we enlarged the red points to facilitate observation. As shown in Fig. 8(a), without the CFCL module, the distribution of the two types of feature points is quite scattered and heavily overlapped, with many red feature points mixed among the blue feature points. In contrast, as shown in Fig. 8(b), the clustering degree of unchanged feature points significantly improves when the CFCL module is added, with all unchanged feature points on the left-hand side. Although some overlap remains on the right-hand side, there is a substantial improvement compared to Fig. 8(a). Therefore, the CFCL module exhibits unique advantages in distinguishing and clustering different features within the domain distribution. It effectively enhances the clustering of similar feature points and reduces confusion between different feature points.

As shown in Fig. 9, we randomly select 51 pairs of bitemporal images from three datasets and reduce the dimensionality of their intermediate layer features to 2-D space. Then, we calculate the Euclidean distances between features to capture their relative positions in the feature space. Finally, we perform correlation matrix analysis to investigate the relationships between different features. It is evident that the data distributions of each dataset vary significantly. However, DDRL effectively captures key discriminative information of multiscale features within the domain distribution, performs reconstruction learning, enhances the correlation of intraclass features, and reduces the correlation of interclass features. As a result, DDRL assists in improving the spatiotemporal consistency of feature mining within the domain distribution, thereby reducing the impact of false detections and missed detections caused by pseudochanges.

E. Ablation Analysis

1) Ablation Analysis of the Proposed Modules: To validate the effectiveness of the proposed GAE and CFCL modules, we conducted an ablation analysis on the three datasets, with the results presented in Table IV. It is evident that using only CSwin as the Siamese encoder to extract multiscale change features yields the least favorable results. This is likely due to the presence of numerous pseudochanges in the bitemporal images, which hampers the accurate identification of the domain distribution. When we incorporate the GAE

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TABLE IV
QUANTITATIVE RESULTS OF THE ABLATION ANALYSIS FROM LEVIR-CD, WHU-CD, AND SYSU-CD. OPTIMAL VALUES ARE SHOWN IN **BOLD**

Dataset	Modules		Precision(%)	Recall(%)	F1(%)	mIOU(%)	Kappa(%)
	GAE	CFCL					
LEVER-CD			92.59	91.22	91.90	91.52	90.90
	✓		93.36	91.03	92.18	91.80	91.22
		✓	93.26	91.24	92.24	91.86	91.29
	✓	✓	**93.54**	**91.31**	**92.41**	**92.02**	**91.48**
WHU-CD			**92.69**	88.73	90.66	91.01	90.21
	✓		92.42	90.00	91.19	91.48	90.76
		✓	92.67	89.96	91.30	91.57	90.87
	✓	✓	92.30	**90.58**	**91.43**	**91.70**	**91.02**
SYSU-CD			**89.74**	73.93	81.07	79.15	75.94
	✓		87.85	75.12	81.33	79.28	76.12
		✓	89.16	76.10	82.11	80.06	77.15
	✓	✓	87.09	**78.16**	**82.38**	**80.19**	**77.32**

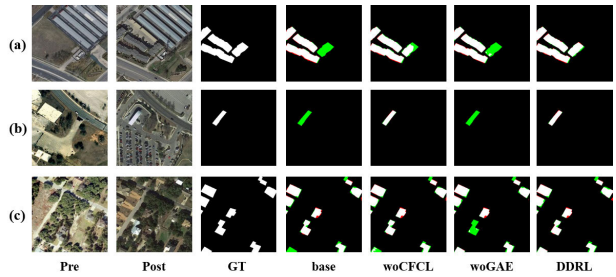


Fig. 10. Visual results of the ablation analysis.

module to significantly enhance low-level domain distribution features, the distribution of low-level domain features becomes more focused on the actual change regions. This leads to a substantial improvement in the recall metric on the WHU-CD and SYSU-CD, reaching 90.00% and 75.12%, respectively. Upon adding the CFCL module, the kappa metric on the three datasets improved significantly, increasing by 0.39%, 0.66%, and 1.21% compared to the baseline model. This indicates that the CFCL module effectively reconstructs learning of high-level fused features, resulting in more compact intraclass features and more dispersed interclass features within the domain distribution. Finally, when we integrate both the GAE and CFCL modules to construct the DDRL framework, the CD results are impressive, with F1 values of 92.41%, 91.43%, and 82.38% on the three datasets, respectively, indicating that DDRL can effectively distinguish potential pseudochange within the domain distribution.

As shown in Fig. 10, we select three different scenarios to illustrate the role of the GAE module and the CFCL module in DDRL and the impact of their interactions on detection results. It is obvious that the CD results of the base model possess more misses and false detections relative to the other three models due to factors such as color differences and occlusions. As shown in Fig. 10(a) and (b), the addition of the GAE module enhances the characterization of the features to some extent, with fewer green areas in the detection results. Moreover, when combined with the CFCL module, DDRL achieves even better CD accuracy, indicating that the contrastive learning strategy effectively leverages the graph-structured features extracted by the GAE module. This synergy enhances the differentiation between

TABLE V
PARAMETER SENSITIVITY ANALYSIS ON THE TEMPERATURE PARAMETER τ FROM LEVIR-CD, WHU-CD, AND SYSU-CD. OPTIMAL VALUES ARE SHOWN IN **BOLD**

Dataset	Temp	F1(%)
LEVER-CD	0.1	**83.65**
	0.5	83.42
	1	83.54
WHU-CD	0.1	**91.43**
	0.5	91.31
	1	91.25
SYSU-CD	0.1	**82.38**
	0.5	82.20
	1	81.96

different categories and regions, significantly reducing noise and pseudochange interference.

2) Parameter Sensitivity Analysis: As shown in Table V, we conducted a sensitivity analysis of the temperature parameter τ in L_{dcl}. The experimental results indicate that a lower temperature parameter ($\tau = 0.1$) makes the model more sensitive to distribution differences within the change domain. This increased sensitivity allows the DDRL to detect subtle changes by highlighting difficult-to-distinguish change regions. As the temperature rises, however, CD performance decreases noticeably, particularly in the SYSU-CD dataset. For instance, when $\tau = 1$, the F1 value drops to 81.96%. This decline occurs because a higher temperature tends to blur the contrast between positive and negative samples, weakening the model's ability to differentiate critical features. Therefore, we ultimately set the temperature parameter τ to 0.1.

3) Coefficients of the Hybrid Loss Function: To investigate the effectiveness of the proposed domain distribution-based CD framework, we set the custom parameter λ in the hybrid loss function from 0 to 1 at intervals of 0.1 and conduct experiments. The radar maps of quantitative results are shown in Fig. 11. We can clearly observe that on the three datasets, DDRL achieves the best CD performance when the custom parameter $\lambda = 0.2$. Our hypothesis is that when $\lambda = 0.2$, $L_{\text{seg}}()$ ensures that the DDRL is sensitive to spatial details of the change regions, while $L_{\text{dcl}}()$ encourages a deeper understanding of contextual and semantic differences. Therefore, this synergy allows DDRL to more accurately capture and

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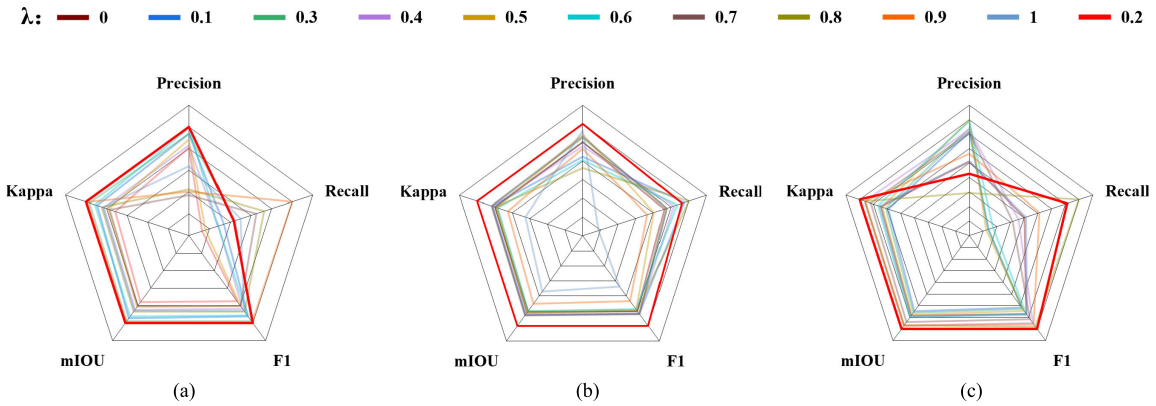


Fig. 11. Radar maps of the quantitative results from (a) LEVIR-CD, (b) WHU-CD, and (c) SYSU-CD.

TABLE VI
EFFICIENCY COMPARISON RESULTS OF DIFFERENT METHODS

Method	Params (M)	FLOPs (G)
FC-EF [38]	1.35	3.58
LUNet [39]	8.45	17.33
STANet [36]	12.21	12.57
IFNet [40]	35.99	82.26
BITNet [41]	3.01	88.50
HFANet [42]	61.99	129.12
DPCCNet [43]	13.74	33.93
WNet [44]	42.76	18.88
CSINet [45]	51.96	324.29
DDRL	31.18	42.08

represent the features of change regions, thereby improving overall CD performance. For other parameter settings, the results are relatively stable, indicating that DDRL exhibits reasonable robustness under various conditions. As a result, we ultimately set the custom parameter λ to 0.2, which not only optimizes performance but also ensures the stability and consistency of DDRL across a wide range of scenarios.

F. Efficiency Analysis

As shown in Table VI, we calculate two metrics for different methods: the number of model parameters (Params) and the number of floating point operations (FLOPs) to measure computational efficiency, respectively. FC-EF, LUNet, and BITNet have relatively small Params due to their relatively simple structure. However, these methods have limited feature extraction capabilities and also perform relatively poorly in F1 metrics. It is worth noting that our proposed DDRL is significantly lower than mainstream methods such as HFANet, WNet, and CSINet in terms of Params, which suggests that DDRL can achieve inspiring CD performance while maintaining a relatively lightweight network structure. In terms of FLOPs, methods such as IFNet, BITNet, HFANet, and CISNet are also much higher than DDRL. Comprehensive efficiency comparison analysis and experimental results show that DDRL based on GAE and CFCL modules can not only effectively overcome the effects of pseudochanges in BCD but also require only a relatively small number of computational resources.

V. CONCLUSION

By analyzing the strengths and weaknesses of segmentation- and metric-based CD methods, we propose the DDRL framework for accurate binary CD, which aims to mitigate the problem of pseudochange interference in bitemporal images. Specifically, we employ a Siamese cross-window self-attention module to extract multiscale features, achieving domain distribution transformation of the original space. Subsequently, we utilize the GAE module to significantly enhance low-level domain distribution features, focusing on changing regions and reducing the interference of ineffective noise. In addition, we design the CFCL module to perform reconstruction learning on high-level fused features, reducing the metric distance between unchanged features in the domain distribution while increasing the metric distance between changed features. The deep contrastive learning loss serves as a pivotal component, constraining DDRL to refine its feature representation learning within the Siamese encoder. By iteratively optimizing with the hybrid loss, the CD results gradually converge with the actual change labels, thereby enhancing CD performance. Experimental results indicate that the proposed DDRL achieves encouraging visual and quantitative results across three public datasets, demonstrating that DDRL can effectively combine the advantages of segmentation- and metric-based CD methods. Although DDRL performs reasonably well at mitigating pseudochange interference, it may still be sensitive to minor changes under extreme conditions, which will require further optimization and adjustment to enhance its robustness and applicability. We hope this study can lay the foundation for future research into implicit or explicit optimization of CD. Future work will focus on enhancing the robustness of the DDRL framework by incorporating multimodal data, such as combining various modes of images and textual information, to capture a broader range of features. This multimodal approach can help reduce the model's sensitivity to minor changes under extreme conditions, thereby improving overall CD performance and generalizability.

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